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Version	Date	Authors	Summary of Changes
2.0	21 Nov 2022	Abhirami, Nobaiha, Sejal, Sayed	Changed one of the design patterns, added justification for the change,updated the class diagrams and added justification for the component diagram.

Document Sign-Off

Name (Position)	Signature	Date

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1 Introduction

1.1 Purpose

This application is a data visualization tool. It allows users to generate different types of graphs (pie chart, line chart, bar chart, area chart) for any selected analysis type.

It is a secure application that prompts users to first log in with a username and password. In case the account does not exist or the password does not match the username, then access is denied. The user first selects the country for which they want to generate graphs from a drop down. The system checks if data fetching for the particular country is allowed. The application provides the user a choice between 8 different types of analysis. The user then enters the start and end year of the data to be visualized. Certain types of analysis are not available for certain years, the system will prompt the user to choose another time frame if entered time frame is not suitable. The user can add different types of viewers (graphs). Certain types of viewers are not compatible with certain analysis, if not compatible user will be directed to choose another viewer. Once the "recalculate" button is clicked, the viewers will provide the analysis of the data.

The application is designed to be improved upon. In the future different types of analyses could be deployed as standalone services accessible over the networks. The UI could also be customizable in the future.

1.2 Overview

The document first provides sequence diagrams based on which the code implementation will be carried out. Major design decisions have been talked about. The architecture of the system is then described. The application uses Observer pattern, Strategy pattern and Factory pattern. The details of how these patterns are used are described. The activities plan shows how the entire project was broken down into smaller digestible tasks. The Gantt chart shows which portions of the project were assigned to which team members, and time allocated for each of the small portions. The group meeting log describes the different meetings that the team had. Every meeting didn't have every team member. We had regular weekly meetings and also meetings on a need-to basis. 16 test case descriptions are added to illustrate how we plan to test the system.

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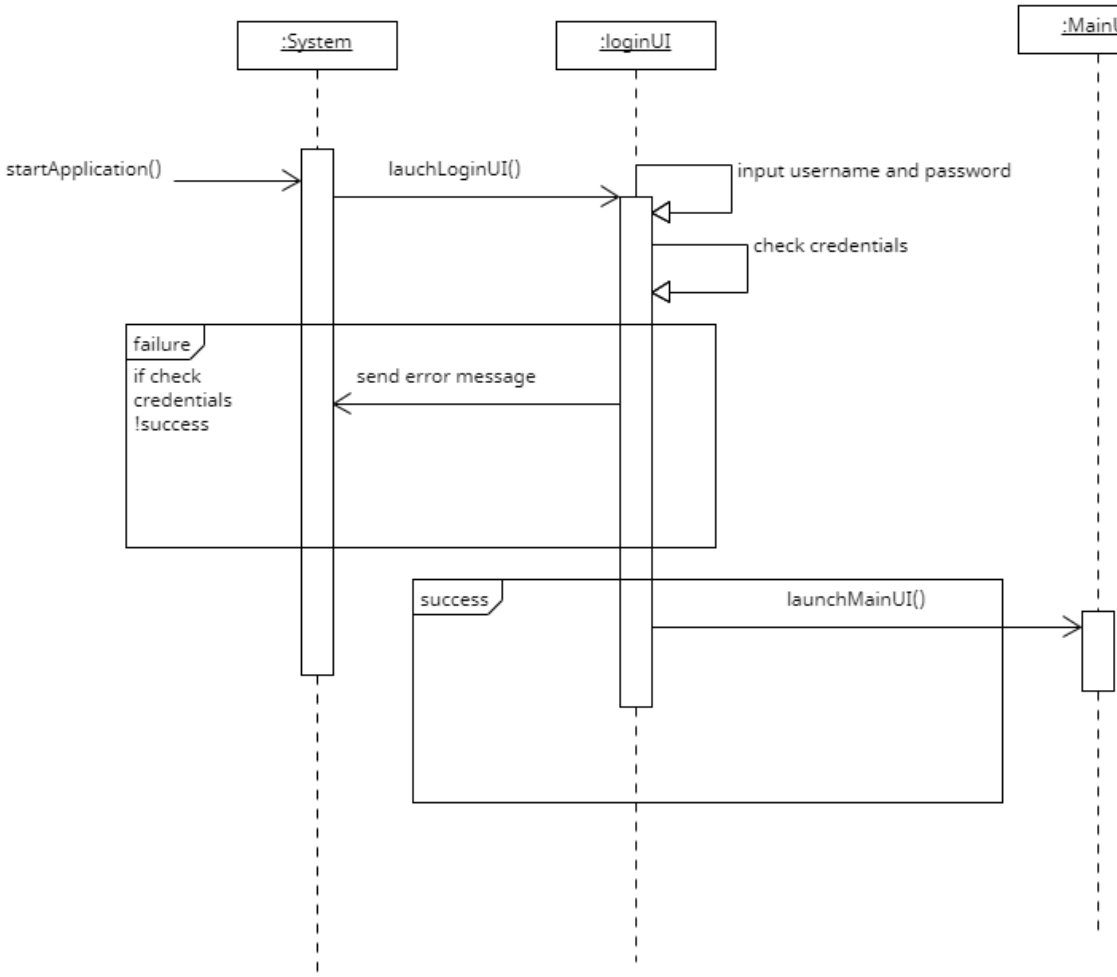
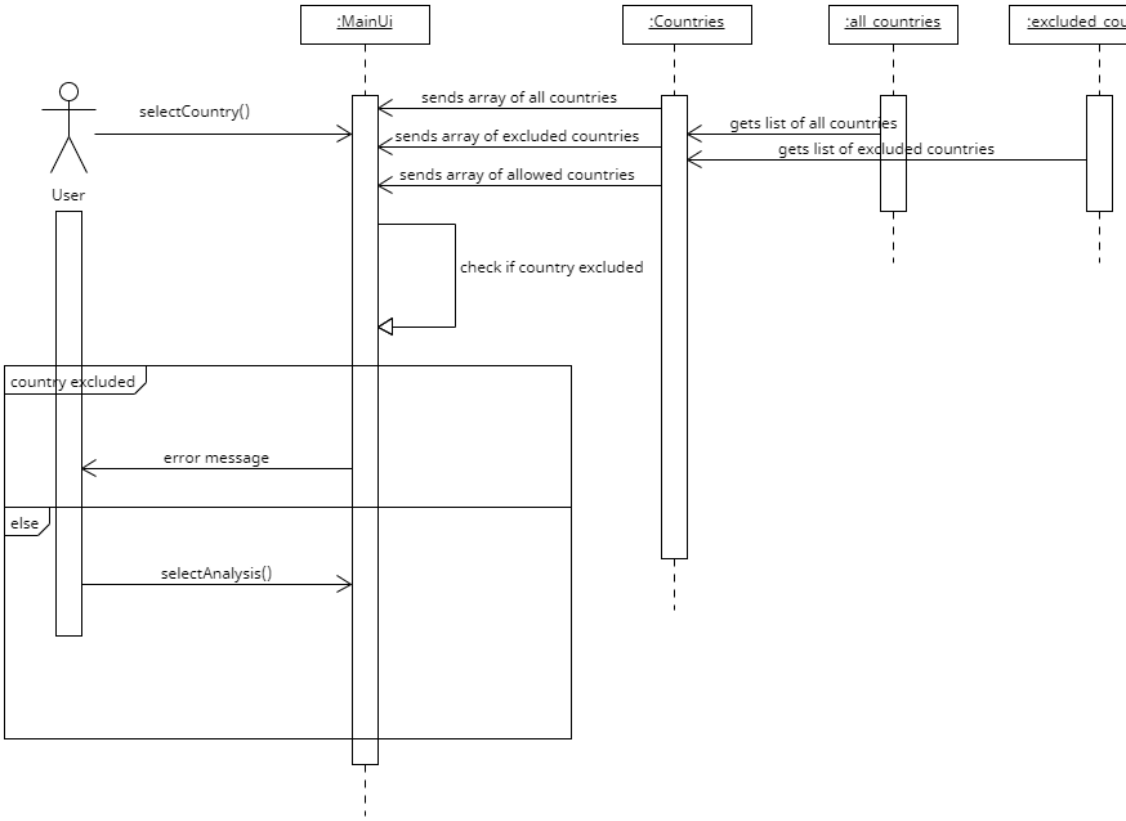
2 Sequence Diagrams

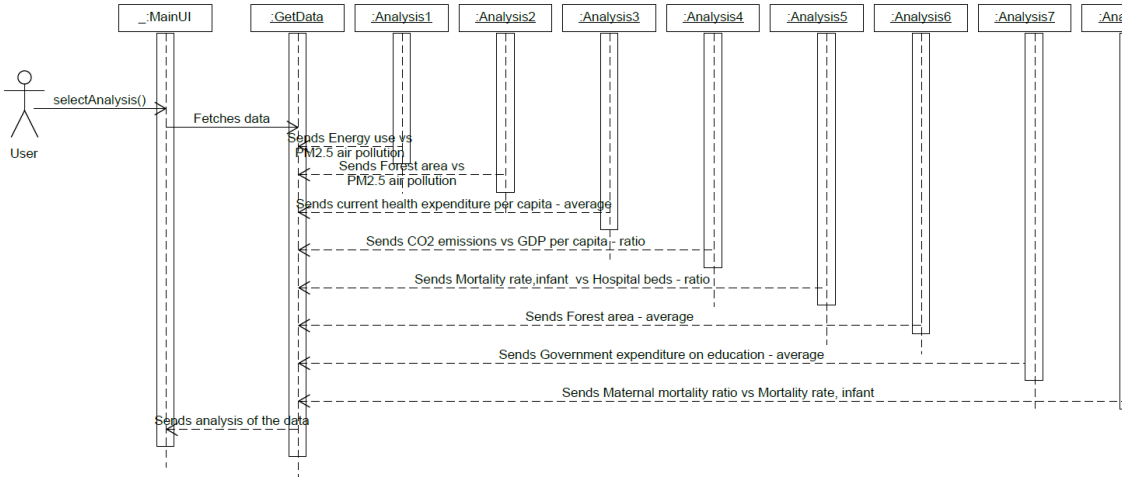
Use Case 1



Use Case 2



Use Case 3



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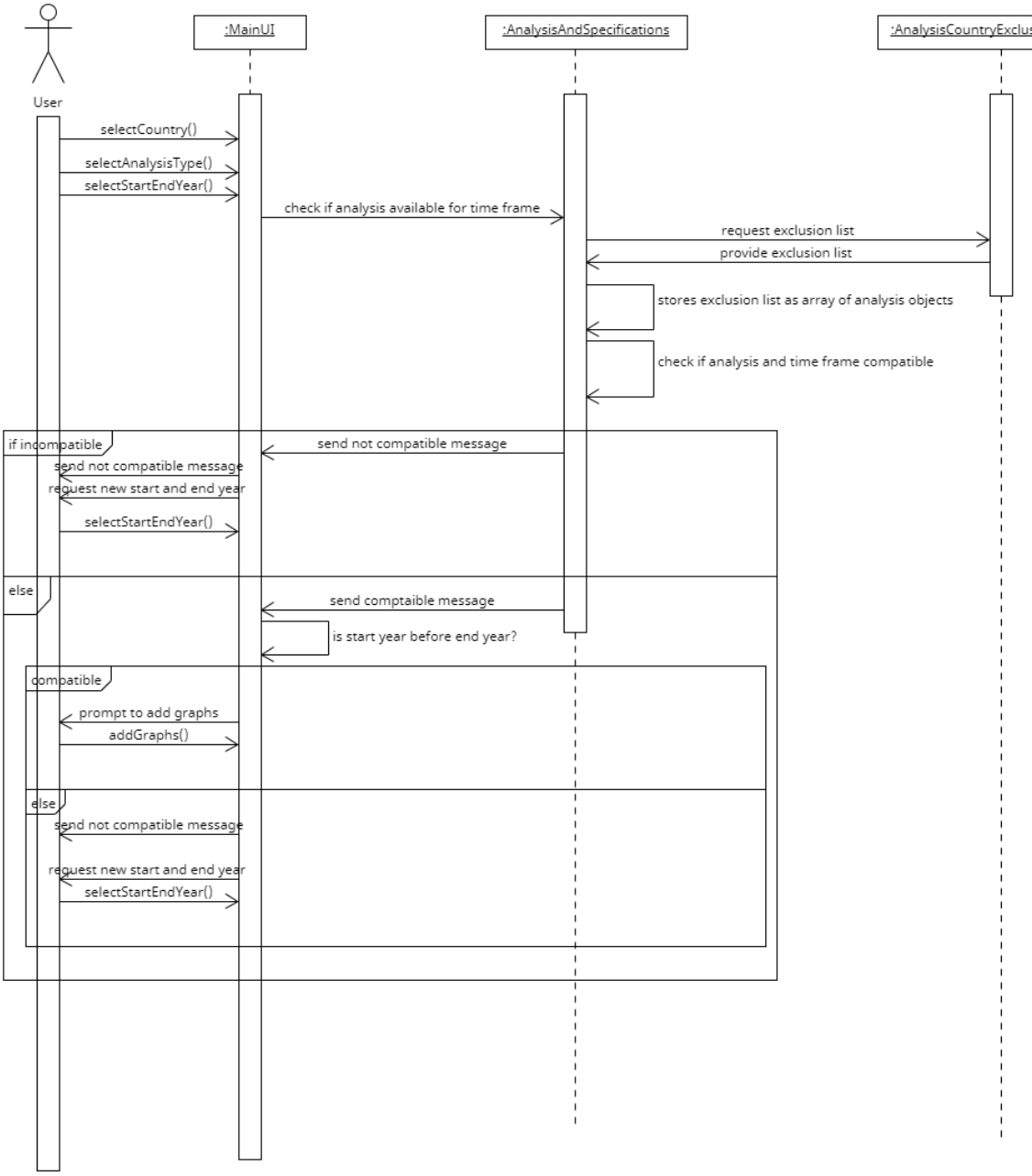
Use Case 4



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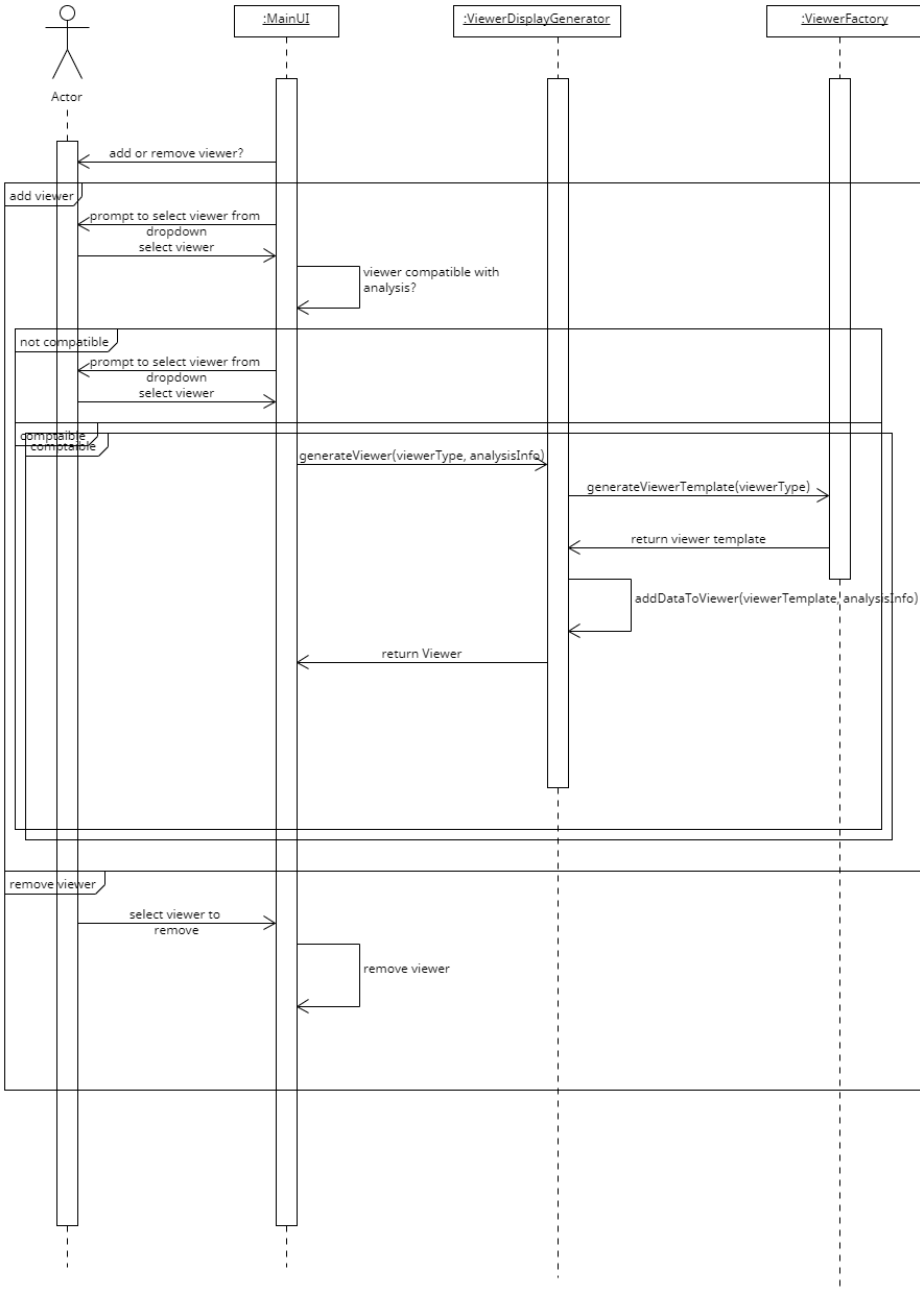
Use Case 5 and 6



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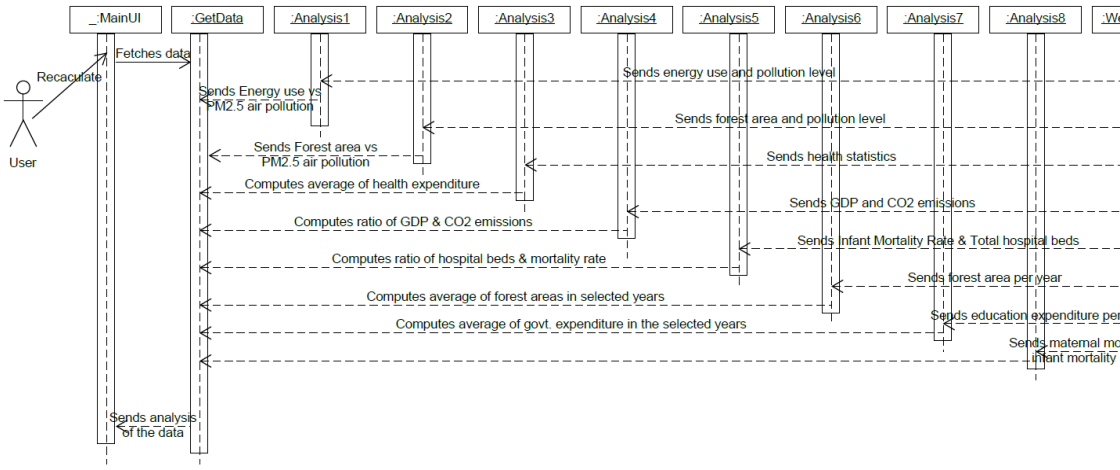
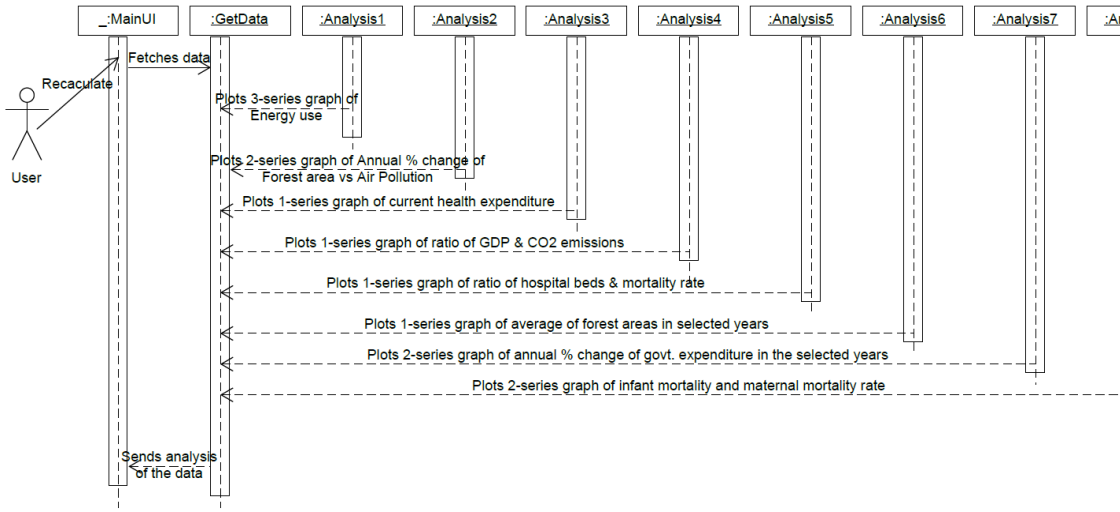
Use Case 7



Use Case 8



3 Architecture

Our design mainly involves

1. MainUI provides the functionality of graph creation for the Analysis
2. IDandPasswords provides password to MainUI.
3. LoginUI provides login information to MainUI.

4., 5., 6. PopupMessages provides alerts for GetData, Restrictions, Analysis.

7. Viewer creates graph for MainUI.

8. Observer pattern provides Observer design functionality to MainUI.

9. IDisplay provides the functionality of displaying a line for MainUI.

10. Restrictions provide the functionality of checking restrictions for GetData (for example: restricted years, countries)

11. Factory provides Factory Design functionality to IDisplay.

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12. Analysis performs analysis according to GetData.

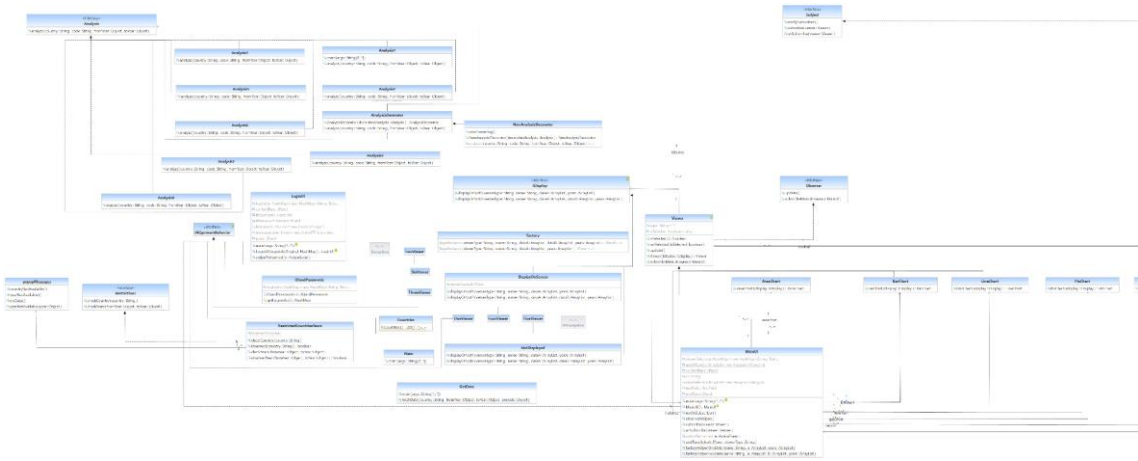
Component Diagram

https://drive.google.com/file/d/1fBYorlrPkcvNbmt61IxnnTsh7QwybFxi/view?usp=share_link

4 Detailed Class Diagrams

a. UML Class Diagrams

https://drive.google.com/file/d/1hdC5t6zq8s4RopGSSaAJx-a4_ocKLy0X/view?usp=share_link



5 Use of Design Patterns

1. Observer Pattern – This pattern will be used to generate the appropriate viewers based on user selection.ObserverPattern package has 3 classes Observer.java, Subject.java and Viewer.java. Here the subject is our MainUI and the observers are the different viewers. Each viewer has an IDisplay behavior which allows us to implement the Strategy pattern.

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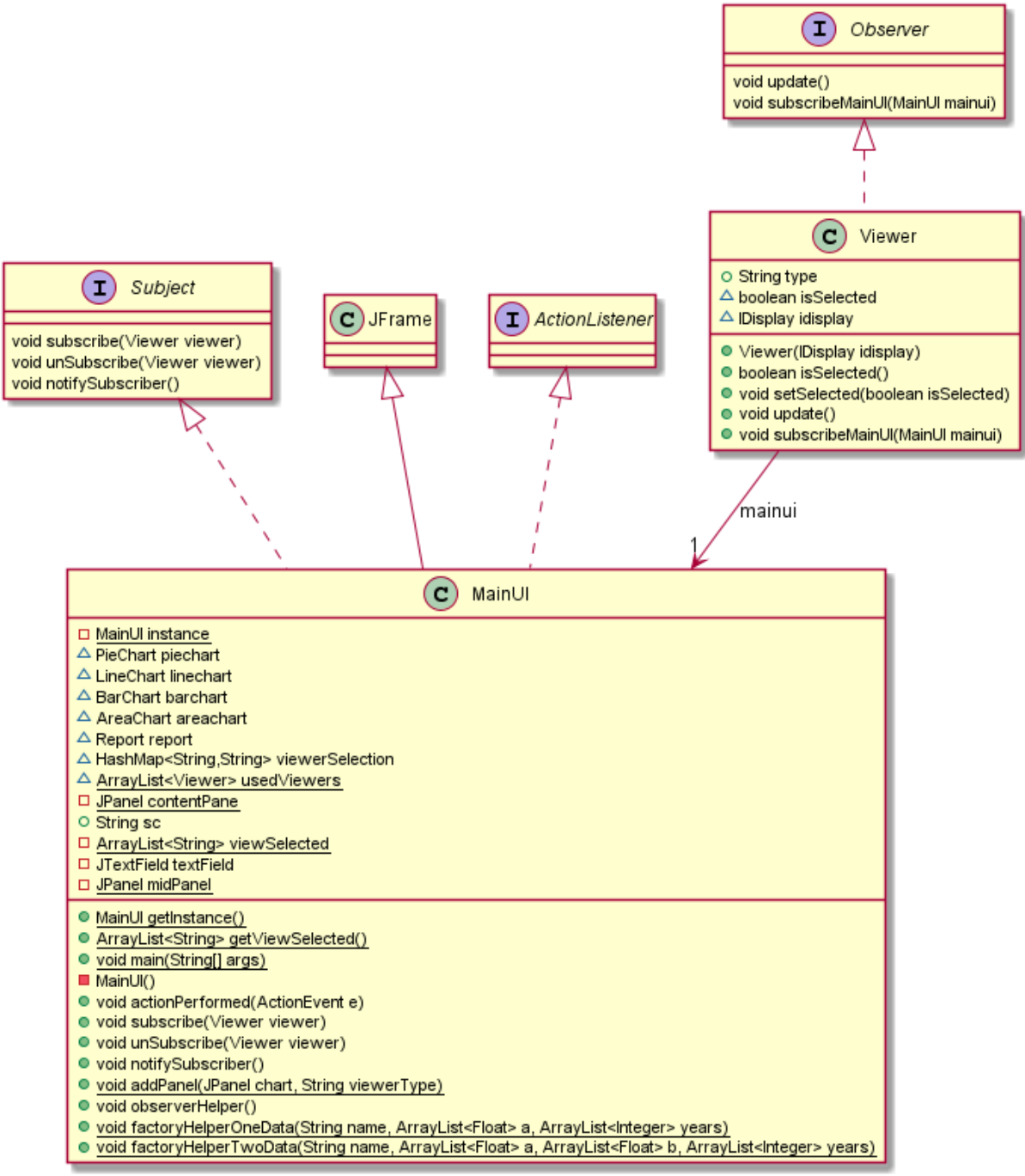
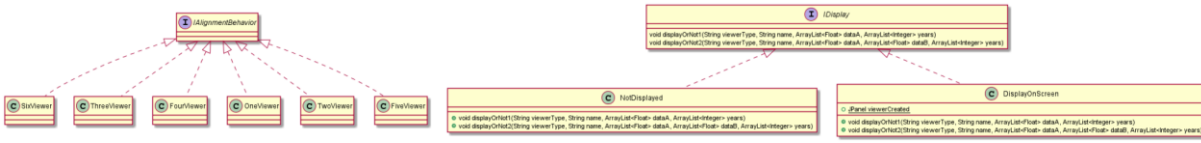


2. Strategy Pattern - This pattern will be used to display only the viewers that are selected. The Strategy package has an interface called IDisplay that is used to model the 2 different display behaviors of the viewers (to be displayed on screen or not). DisplayOnScreen.java displays the selected viewers using JFree charts created by the Factory Pattern.



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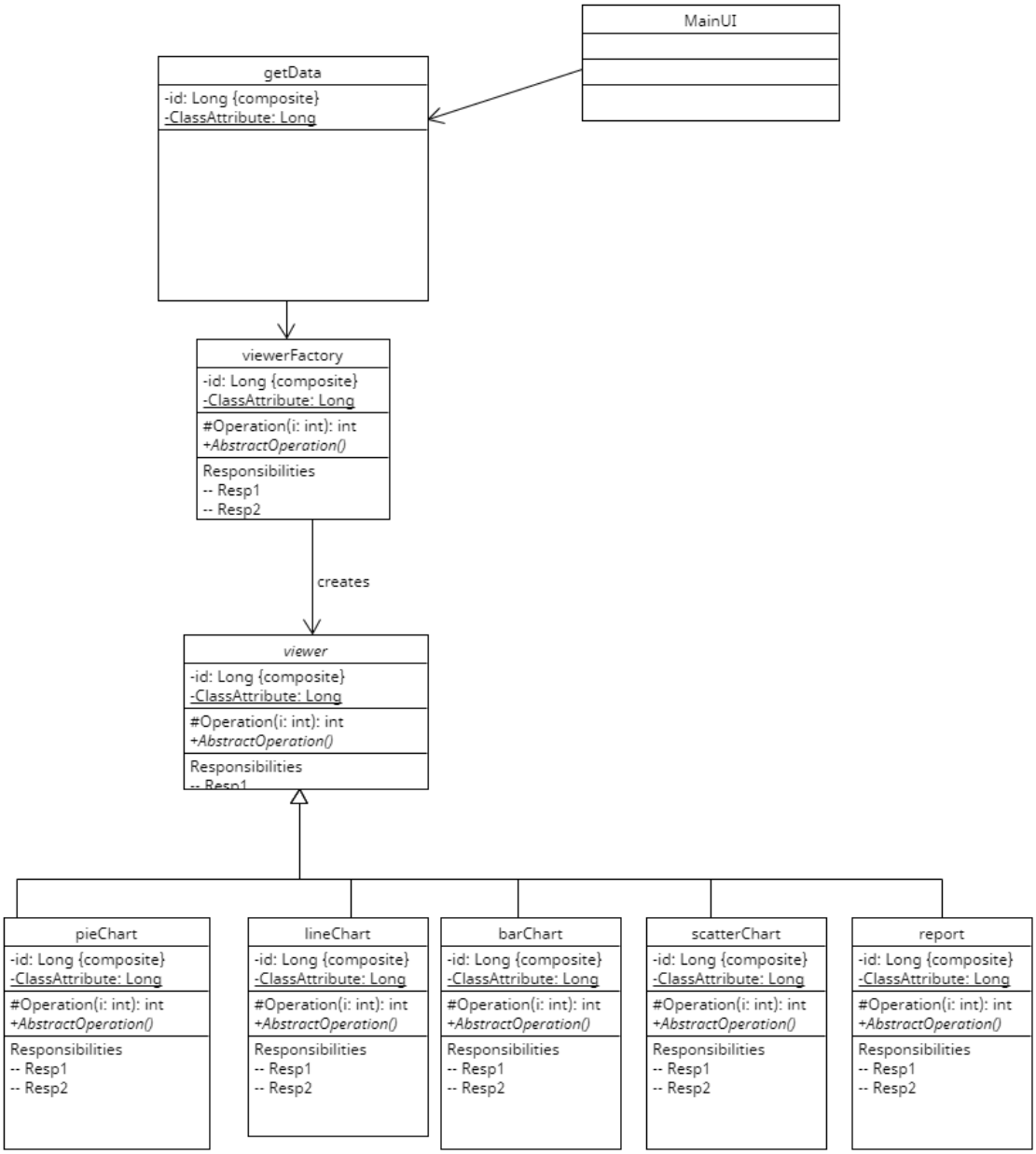
3. Factory Pattern – This pattern will be used to create different kinds of viewers. Users have the option of using 5 different kinds of viewers namely: pieChart, lineChart, barChart, report and scatterChart. ViewerFactory creates the correct viewer based on what the user selects at run-time.



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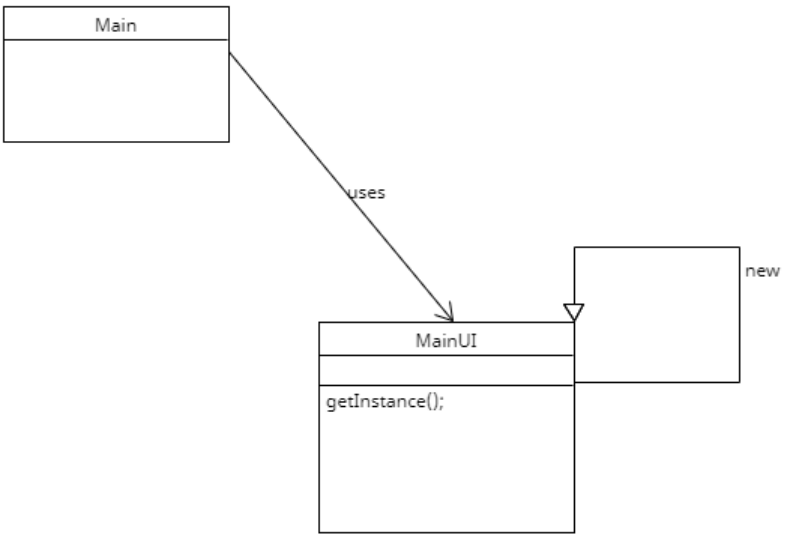
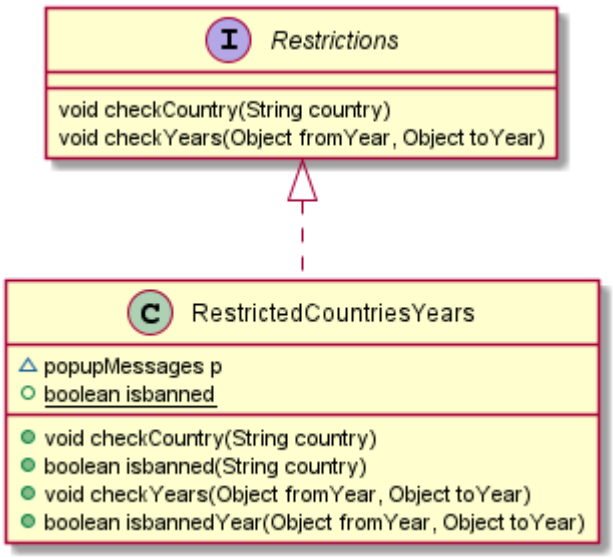
4. Singleton Pattern - This pattern is used to ensure that only one instance of the Main UI is created.



5. Proxy Pattern - In proxy pattern, a class represents functionality of another class. This type of design pattern falls under structural pattern. It is used when we want to provide controlled access of a functionality.

   For our project we are using proxy pattern to check if the selected country and years are banned or not. The proxy interface is called Restrictions.java & proxy class is called RestrictedCountriesYears.java. The proxy class is called from the GetData.java class. When the user selects all the attributes and clicks on the recalculate button the GetData class gets the attributes and with the help of proxy pattern checks if the selected country & years are available. If they are available the data is displayed, if not a pop up message is shown to the user.



6. Decorator Pattern - Decorator Pattern allows a user to add new functionality to an existing object without altering its structure. For future implementations our project should be able to add new analysis. For which we chose to use the Decorator Pattern. We

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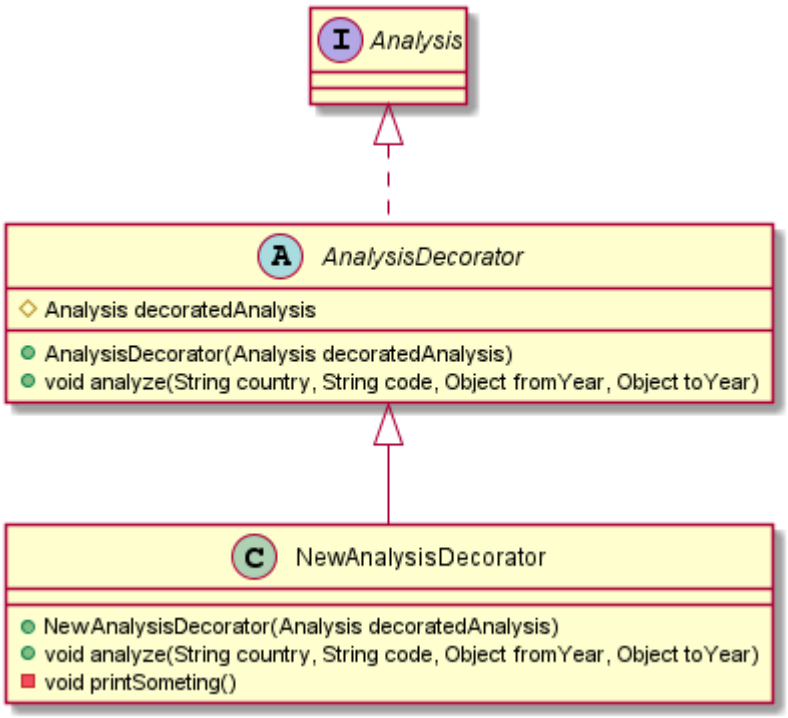
have an abstract decorator class AnalysisDecorator.java and concrete decorator class NewAnalysisDecorator.java.



2. 6. Justification for design pattern change

Previously, we had decided to use the Builder Pattern to assemble the different viewers within the Main UI. In this idea, the product would have been the main viewer's display area. However, after we started implementing the project we realized that this portion of the project was being implemented by the Observer pattern and Strategy pattern. Thus we decided to omit the Builder Pattern.A new addition is the Decorator pattern to make the application extendable in the future.

3. 7. Activities Plan

a. 7.1 Development Tasks and Their Priorities

4. Create GUI for login page. [Priority - Medium]
5. Credential verification for login page. [Priority - Medium]
6. Figure out which design patterns we are using, and how to use them in our project. [Priority - High]
7. Create sequence diagrams for use case 1, 3, 4 and 8. [Priority - Medium]
8. Create sequence diagrams for use case 2, 5, 6 and 7. [Priority - Medium]
9. Create a list of countries for which data fetching is not allowed. [Priority - Low]
10. Fetch data from a given website, understand the basic way to do it. [Priority - High]
11. Create stand alone classes for 8 different types of analyses. [Priority - High]
12. Fetch data for the stand alone classes for different types of analysis. [Priority - Medium]
13. Create basic structure for main UI. [Priority - Medium]
14. Connect the login page and main page. [Priority - Low]

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15. Create UML diagram. [Priority - Medium]
16. Write Test case descriptions. [Priority - Medium]
17. Implement Observer Pattern. [Priority - Medium]
18. Implement Strategy Pattern. [Priority - Medium]
19. Implement Singleton Pattern. [Priority - Medium]
20. Implement remaining design patterns. [Priority - Medium]
21. Ensure that every time recalculate button is pressed new set of graphs are being displayed. [Priority - Low]
22. Connect JFree charts to the display area of the MainUI. [Priority - High]
23. Create the component diagram. [Priority - High]
24. Write down justifications for the component decompositions and interactions. [Priority - Medium]
25. Create updated UML diagram. [Priority - High]
26. Write justification for choosing the six design patterns and justifications for design pattern choice changes compared to Deliverable 1. [Priority - Medium]

a. 7.2 A GANTT Diagram for Planned Activities

https://docs.google.com/spreadsheets/d/1OlpICk9SPh6klQk0BsoTGYQ998OQBUsj/edit?usp=share_link&ouid=104320405390679160739&rtpof=true&sd=true

b. 7.3 Group Meeting Logs

Present Group Members	Meeting Date	Issues Discussed / Resolved
Abhirami, Sayed, Sejal, Nohaiha	Sep 21 2022	First meeting Decided to read through all the use cases Read through instructions together
Sayed and Sejal	Sep 25, 2022	Sets up maven and github for project
Abhirami, Sayed, Sejal, Nobaiha	Sep 28, 2022	Sayed decides to develop the GUI Nobaiha and Sejal decide to work on data fetching Abhirami decides to collaborate with Sayed and works on design patterns to be included in project Creates Gantt chart
Nobaiha and Sejal	Oct 4, 2022	Debugging session for Data fetching
Sayed and Abhirami	Oct 11, 2022	Integrating main UI and login UI Discussed about how to start working on analysis classes

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Abhirami and Sejal	Oct 11, 2022	Abhirami updates on progress with UI and data analysis Sejal updates on progress with data fetching
Abhirami, Nobaiha, Sejal, Sayed	Oct 12, 2022	Discuss how to use data fetching in analysis classes Discuss equations to use for analysis
Abhirami, Nobaiha, Sejal, Sayed	Nov 2, 2022	Read the instructions for Deliverable 2.
Abhirami, Nobaiha, Sejal, Sayed	Nov 4, 2022	Made a list of to-do's and divided work.
Abhirami, Sayed, Sejal	Nov 8, 2022	Completed 50% of the implementation of Observer pattern,
Abhirami, Nobaiha	Nov 8, 2022	Discussed changing one of the design patterns.
Abhirami, Sayed	Nov 12, 2022	Completed the implementation of Observer pattern and 50% of Strategy pattern.
Abhirami, Nobaiha, Sejal	Nov 18, 2022	Worked on Factory pattern together.
Sejal, Sayed	Nov 19, 2022	Completed component diagram. Wrote justifications for component decomposition and interactions.
Abhirami, Sayed, Sejal	Nov 18, 2022	Worked on implementing JFree chart for the Factory pattern.
Nobaiha, Sayed	Nov 20, 2022	Finished up implementation.

27. 8. Test Driven Development

Test ID	Abhirami - 1
Category	Entering username and password. Checking if the account exists.
Requirements Coverage	UC1- The user logs into the system
Initial Condition	Login page appears
Procedure	The user provides an incorrect user name 2. The user provides a password 3. The user presses "Login" button.

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Expected Outcome	Error message “Username not found” is displayed.

Test ID	Abhirami - 2
Category	Evaluation of user credentials
Requirements Coverage	US1- User logs into system
Initial Condition	Login UI appears, allows to user to type in username and password
Procedure	1. The user provides a correct user name 2. The user provides an incorrect password 3. The user presses the “login” button
Expected Outcome	Error message of “Wrong credentials” displayed.

Test ID	Abhirami-3
Category	Evaluation of user credentials
Requirements Coverage	US1- User logs into system
Initial Condition	Login UI appears, allows to user to type in username and password
Procedure	1. The user provides a correct user name 2. The user provides the correct password 3. The user presses the “login” button
Expected Outcome	Message “login successful” appears, then Main UI page displayed.

Test ID	Abhirami-4
Category	Checking if the analysis and country selected is compatible
Requirements Coverage	US4 - Selecting the years for which analysis type is to be performed
Initial Condition	The system has been initiated and runs
Procedure	1. The user is able to login successfully and moves onto the Main UI.

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	2. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 3. The analysis and year chosen are incompatible. 4. Viewer selected. 5. User clicks “Recalculate” button.
Expected Outcome	Message “Analysis type not available o

Test ID	Nobaiha -1
Category	Choosing a country and checking if data can be fetched for that country
Requirements Coverage	UC2-Selecting a country to fetch and visualize data for
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The country they chose is on the list for which data fetching is not allowed. 7. The user clicks on the recalculate button.
Expected Outcome	The console prints “Sorry cannot fetch data for this country”
Notes	In order to get the expected outcome, the user needs to choose a country that is on the list. (For example: Algeria, Brunei , Bosnia and Herzegovina etc.)

Test ID	Nobaiha -2

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Category	Checking if the start year is before end year
Requirements Coverage	UC4- Selecting the years for which the analysis type is to be performed
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The start year they chose is after the end year. 7. The user clicks on the recalculate button.
Expected Outcome	The console prints "Can't fetch data as the start year is after the end year."
Notes	In order to get the expected outcome, the user needs to choose a start year that is after the end year.

Test ID	Nobaiha -3
Category	Checking if the years selected satisfies the requirements
Requirements Coverage	UC4- Selecting the years for which the analysis type is to be performed
Initial Condition	The system has been initiated and runs

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Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The end year they chose is on the list of years not available for certain analysis. 7. The user clicks on the recalculate button.
Expected Outcome	The console prints "Analysis not available for the year 2021."
Notes	In order to get the expected outcome, the user needs to choose an end year that is on the list of unavailable years for certain analysis.

Test ID	Nobaiha -4
Category	Performing an analysis
Requirements Coverage	UC7 - Performing the analysis
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The user clicks on the recalculate button
Expected Outcome	The console prints "The average forest area of Canada over the selected years is 38.727955"

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Notes	In order to get the expected outcome, the user needs to choose *Canada* for the country, *2010* for start year , *2015* for end year & *Forest area - average* for analysis method.

Test ID	Sejal-1
Category	Performing an analysis
Requirements Coverage	UC7 - Performing the analysis
Initial Condition	The system has been initiated and runs. The user has selected the country, from year, to year, desired analysis
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The user clicks on the recalculate button
Expected Outcome	"Energy use of India in 2015 is 0.0 PM2.5 air pollution of India in 2015 is 89.30334 Energy use of India in 2014 is 636.57184 PM2.5 air pollution of India in 2014 is 89.62238 Energy use of India in 2013 is 605.79407 PM2.5 air pollution of India in 2013 is 91.804794 Energy use of India in 2012 is 599.15564 PM2.5 air pollution of India in 2012 is 88.16944 Energy use of India in 2011 is 577.99445 PM2.5 air pollution of India in 2011 is 97.59938" It gives the output for energy use & PM2.5 air pollution in India for the year 2010-2015.
Notes	In order to get the expected outcome, the user needs to choose *India* for the country, *2011* for start year , *2015* for end year & *Energy use vs PM2.5 air pollution* for analysis method.

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Test ID	Sejal-2
Category	Displaying the analysis
Requirements Coverage	UC8 - Displaying the analysis
Initial Condition	The system has been initiated and runs. The user has selected the country, from year, to year, desired analysis
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The user clicks on the recalculate button
Expected Outcome	The selected visualisation of the desired type of analysis would display for the country selected in the chosen years.
Notes	

Test ID	Sejal-3
Category	Adding and removing visualisation graphs
Requirements Coverage	UC5,6 - Adding and removing visualisation graphs
Initial Condition	The system has been initiated and runs.
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched.

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	6. The user clicks on the recalculate button 7. To add new graphs, the user clicks on the "+" button and selects new graph to be added.
Expected Outcome	The user is able to add selected type of graph for the chosen analysis.
Notes	

Test ID	Sejal-4
Category	Selecting the years for which the analysis type is to be performed
Requirements Coverage	UC4 - Selecting the years for which the analysis type is to be performed
Initial Condition	The system has been initiated and runs.
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The user clicks on the recalculate button.
Expected Outcome	Analysis not available for the year 2020
Notes	In order to get the expected outcome, the user needs to choose *Algeria* for the country, *2020* for start year , *2021* for end year & *CO2 emissions vs GDP per capita - ratio* for analysis method.

Test ID	Mujtaba -1
Category	Displaying multiple data category

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Requirements Coverage	UC5&6-Adding and Removing visualization graphs to display the obtained/computed data
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched and the available view type for that data. 6. The user clicks on the "+ (Plus)" button then the recalculate button. 7. The user chooses different kind of analysis and view type and follows step 6 again.
Expected Outcome	(*Future Implementation*) Two graphs are shown in the viewer.
Notes	

Test ID	Mujtaba -2
Category	Checking if the graph can be displayed for the analysis
Requirements Coverage	UC5&6-Adding and Removing visualization graphs to display the obtained/computed data
Initial Condition	The system has been initiated and runs

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Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched and the available view type for that data. 6. The user clicks on the "+ (Plus)" button then the recalculate button.
Expected Outcome	(*Future Implementation*) A message is shown that graph for this data type is not available.
Notes	In order to get the expected outcome, the user needs to choose *Pie Chart* from available views and *Forest area vs PM2.5 air pollution* from analysis method.

Test ID	Mujtaba -3
Category	Checking if data required for computation is available
Requirements Coverage	UC7 - Performing the analysis
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched and the available view type for that data. 6. The user clicks on the "+ (Plus)" button then the recalculate button.
Expected Outcome	(*Future Implementation*) A message is shown that data to do this computation is not available.

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Notes	In order to get the expected outcome, the user chooses *Nigeria* for the country, *2010* for start year , *2019* for end year & *Government expenditure on education – average* for analysis method.

Test ID	Mujtaba -4
Category	Performing an analysis
Requirements Coverage	UC7 - Performing the analysis
Initial Condition	The system has been initiated and runs
Procedure	1. The user selects login 2. The user provides a user name 3. The user provides a password 4. The user logs-in into the system and is presented with the main UI window. 5. The User chooses an analysis method , the country and the start & end year of the data that needs to be fetched. 6. The user clicks on the recalculate button
Expected Outcome	Infant Mortality Rate of Spain in 2016 is 2.7 Hospital Bed of Spain in 2016 is 2.97 Ratio of Infant Mortality Rate vs Hospital Bed Spain in 2016 is 1.1 Infant Mortality Rate of Spain in 2015 is 2.7 Hospital Bed of Spain in 2015 is 2.98 Ratio of Infant Mortality Rate vs Hospital Bed Spain in 2015 is 1.1037037 Infant Mortality Rate of Spain in 2014 is 2.8 Hospital Bed of Spain in 2014 is 2.97 Ratio of Infant Mortality Rate vs Hospital Bed Spain in 2014 is 1.0607144
Notes	In order to get the expected outcome, the user chooses *Spain* for the country, *2014* for start year , *2016* for end year *Mortality rate,infant vs Hospital beds - ratio* for analysis method.

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